

Experiment – 3

Aim: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

1. Identify all entity types and entity sets.

The following entity types are identified from the problem statement:

1. **RESEARCHER**
2. **DEPARTMENT**
3. **RESEARCH_PROJECT**
4. **STUDENT**
5. **PUBLICATION**
6. **PROJECT_ASSISTANT**

The corresponding entity sets contain all instances of these entities in the university research system.

2. Identify strong and weak entity sets.

Strong Entity Sets:

- RESEARCHER
- DEPARTMENT
- RESEARCH_PROJECT
- STUDENT
- PUBLICATION

These entities have their own identifying attributes.

Weak Entity Set:

- PROJECT_ASSISTANT

PROJECT_ASSISTANT is a weak entity because an assistant is identified only within the context of a particular research project and does not have a globally unique identifier.

3. Identify the owner entity and partial key of the weak entity.

- **Weak Entity:** PROJECT_ASSISTANT
- **Owner Entity:** RESEARCH_PROJECT
- **Partial Key:** Assistant_Name

Assistant_Name identifies an assistant only within a particular project. The same assistant name may occur in different projects.

Therefore, after conversion into relations, the composite key of PROJECT_ASSISTANT is:

(Project_ID, Assistant_Name)

4. Identify simple attributes.

The simple attributes are:

RESEARCHER:

- Researcher_ID
- Name
- Date_of_Birth
- Email

DEPARTMENT:

- Department_ID
- Department_Name
- Location

RESEARCH_PROJECT:

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget

STUDENT:

- Student_ID
- Name
- Email
- Date_of_Birth

PUBLICATION:

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

PROJECT_ASSISTANT:

- Assistant_Name
- Role
- Joining_Date

The components of Address are also simple attributes:

- Street
- City
- State
- PIN_Code

5. Identify composite attributes.

The composite attribute is:

Address

Address can be divided into:

- Street
- City
- State
- PIN_Code

Therefore:

Address = {Street, City, State, PIN_Code}

6. Identify single-valued attributes.

The following attributes are single-valued:

- Researcher_ID
- Name
- Date_of_Birth
- Email
- Department_ID
- Department_Name
- Location
- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget
- Student_ID
- Title
- Publication_ID
- Publication_Date

- DOI
- Publisher
- Assistant_Name
- Role
- Joining_Date
- Street
- City
- State
- PIN_Code

Age is also single-valued for a particular researcher, but it is a **derived attribute**.

7. Identify multi-valued attributes.

The multi-valued attributes are:

- **Phone_No** of RESEARCHER
- **Keywords** of RESEARCH_PROJECT

A researcher can have multiple phone numbers, and a research project can have multiple keywords.

These are represented using **double ovals** in the ER diagram.

8. Identify stored attributes.

The stored attributes are the attributes whose values are directly maintained in the database.

Examples include:

- Researcher_ID
- Name
- Date_of_Birth
- Email
- Phone_No
- Street
- City
- State
- PIN_Code
- Department_ID
- Department_Name
- Location
- Project_ID

- Project_Title
- Start_Date
- End_Date
- Budget
- Keywords
- Student_ID
- Title
- Publication_ID
- Publication_Date
- DOI
- Publisher
- Assistant_Name
- Role
- Joining_Date

9. Identify derived attributes.

The derived attribute is:

Age

Age is calculated from **Date_of_Birth**.

Therefore:

Age = Current Date – Date_of_Birth

Age is represented using a **dashed oval** in the ER diagram.

10. Identify key attributes.

The key attributes are:

Entity	Key Attribute
RESEARCHER	Researcher_ID
DEPARTMENT	Department_ID
RESEARCH_PROJECT	Project_ID
STUDENT	Student_ID
PUBLICATION	Publication_ID
PROJECT_ASSISTANT	Assistant_Name (Partial Key)

11. Identify candidate keys and select appropriate primary keys.

Entity	Candidate Key	Selected Primary Key
RESEARCHER	Researcher_ID	Researcher_ID
DEPARTMENT	Department_ID	Department_ID
RESEARCH_PROJECT	Project_ID	Project_ID
STUDENT	Student_ID	Student_ID
PUBLICATION	Publication_ID	Publication_ID
PROJECT_ASSISTANT	Project_ID + Assistant_Name	Project_ID + Assistant_Name

For PROJECT_ASSISTANT, Assistant_Name alone is not sufficient because the same name can occur in different projects. Hence, **(Project_ID, Assistant_Name)** forms the composite key after conversion into relations.

Note: DOI can also serve as a candidate key for PUBLICATION only when the database guarantees that every DOI is unique.

12. Identify at least one composite key.

A composite key is:

(Project_ID, Assistant_Name) in the PROJECT_ASSISTANT relation.

Here:

- Project_ID identifies the research project.
- Assistant_Name identifies the assistant within that project.

Together, they uniquely identify a project assistant.

Another composite key can occur in the **WORKS_ON** relation:

(Researcher_ID, Project_ID)

13. Identify appropriate foreign keys after converting the ER model into relations.

After converting the ER model into relations, the foreign keys are:

Relation	Foreign Key
RESEARCHER	Department_ID
STUDENT	Researcher_ID
WORKS_ON	Researcher_ID, Project_ID
PUBLICATION	Project_ID
PROJECT_ASSISTANT	Project_ID

The foreign keys establish relationships between the corresponding relations.

14. Identify all major relationships between the entities.

Relationship	Entities Involved	Cardinality
BELONGS_TO	RESEARCHER – DEPARTMENT	N : 1
WORKS_ON	RESEARCHER – RESEARCH_PROJECT	M : N
GUIDES	RESEARCHER – STUDENT	1 : N
PRODUCES	RESEARCH_PROJECT – PUBLICATION	1 : N
HAS	RESEARCH_PROJECT – PROJECT_ASSISTANT	1 : N

Cardinality explanation:**BELONGS_TO (N : 1)**

Many researchers can belong to one department, while each researcher belongs to one department.

WORKS_ON (M : N)

A researcher can work on multiple research projects, and a research project can involve multiple researchers.

GUIDES (1 : N)

One researcher can have multiple students working under them, while each student works under a researcher.

PRODUCES (1 : N)

One research project can produce multiple publications, while each publication is associated with a research project.

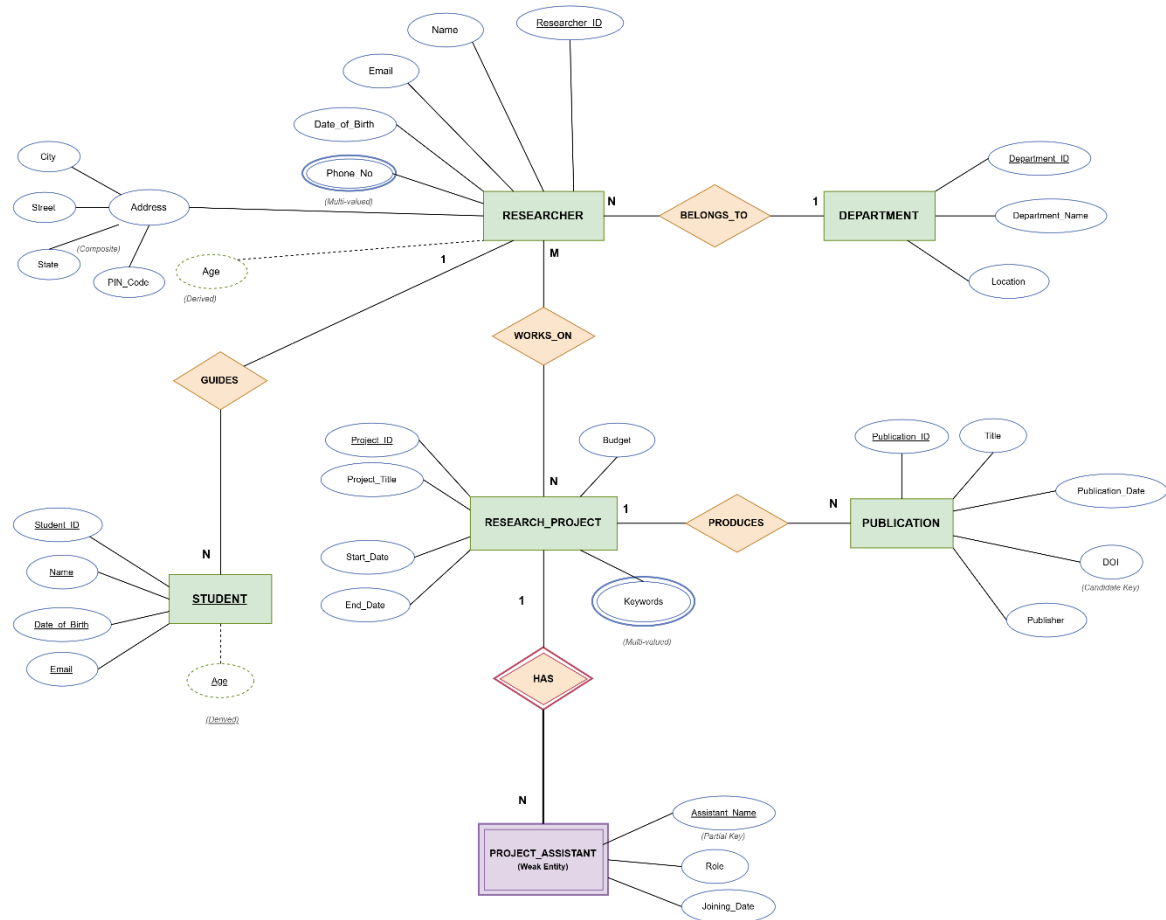
HAS (1 : N)

One research project can have multiple project assistants. PROJECT_ASSISTANT is a weak entity dependent on RESEARCH_PROJECT.

HAS is therefore shown as an **identifying relationship** using a **double diamond**.

15. Draw the complete ER diagram using standard ER notation.

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16. Prepare a table showing each attribute and its attribute type and key status.

Entity	Attribute	Attribute Type	Key Status
RESEARCHER	Researcher_ID	Simple, Single-valued, Stored	Primary Key
RESEARCHER	Name	Simple, Single-valued, Stored	—
RESEARCHER	Date_of_Birth	Simple, Single-valued, Stored	—
RESEARCHER	Age	Simple, Single-valued, Derived	—
RESEARCHER	Email	Simple, Single-valued, Stored	—
RESEARCHER	Phone_No	Simple, Multi-valued, Stored	—
RESEARCHER	Address	Composite, Stored	—
RESEARCHER	Street	Simple, Single-valued, Stored	—
RESEARCHER	City	Simple, Single-valued, Stored	—

RESEARCHER	State	Simple, Single-valued, Stored	—
RESEARCHER	PIN_Code	Simple, Single-valued, Stored	—
DEPARTMENT	Department_ID	Simple, Single-valued, Stored	Primary Key
DEPARTMENT	Department_Name	Simple, Single-valued, Stored	—
DEPARTMENT	Location	Simple, Single-valued, Stored	—
RESEARCH_PROJECT	Project_ID	Simple, Single-valued, Stored	Primary Key
RESEARCH_PROJECT	Project_Title	Simple, Single-valued, Stored	—
RESEARCH_PROJECT	Start_Date	Simple, Single-valued, Stored	—
RESEARCH_PROJECT	End_Date	Simple, Single-valued, Stored	—
RESEARCH_PROJECT	Budget	Simple, Single-valued, Stored	—
RESEARCH_PROJECT	Keywords	Simple, Multi-valued, Stored	—
STUDENT	Student_ID	Simple, Single-valued, Stored	Primary Key
STUDENT	Name	Simple, Single-valued, Stored	—
STUDENT	Email	Simple, Single-valued, Stored	—
STUDENT	Date_of_Birth	Simple, Single-valued, Stored	—
PUBLICATION	Publication_ID	Simple, Single-valued, Stored	Primary Key
PUBLICATION	Title	Simple, Single-valued, Stored	—
PUBLICATION	Publication_Date	Simple, Single-valued, Stored	—
PUBLICATION	DOI	Simple, Single-valued, Stored	Candidate Key*
PUBLICATION	Publisher	Simple, Single-valued, Stored	—
PROJECT_ASSISTANT	Assistant_Name	Simple, Single-valued, Stored	Partial Key
PROJECT_ASSISTANT	Role	Simple, Single-valued, Stored	—
PROJECT_ASSISTANT	Joining_Date	Simple, Single-valued, Stored	—

* DOI is a candidate key only when uniqueness of DOI is guaranteed.